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TIME SERIES ANALYSIS AND ARIMA FORECASTING OF MALAYSIA WEEKLY RETAIL FUEL PRICES (RON95)

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INTRODUCTION

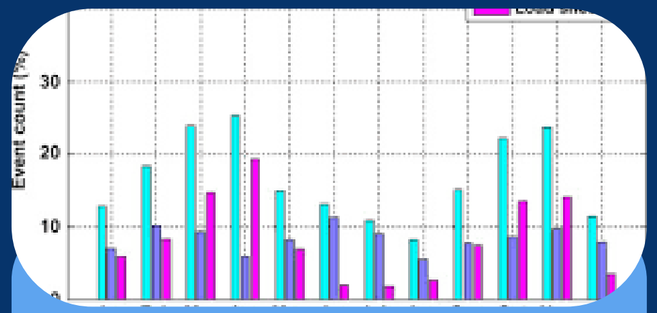
- Fuel prices are critical economic indicators in Malaysia, impacting transportation cost and household expenditures.

- This study examines the historical pattern of a variable over time, identify trend and seasonal components, and build forecasting models.



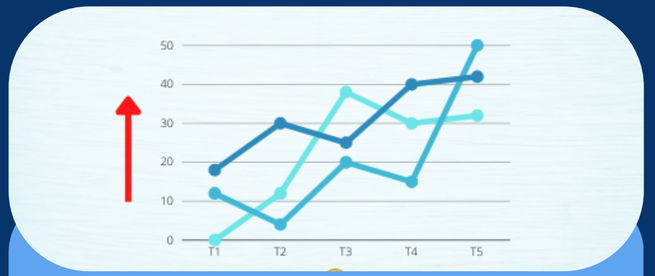
- Accurate analysis and forecasting of retail fuel prices are essential for businesses and consumers to support effective planning.

OBJECTIVES



Trend Analysis

To analyze the historical trends and patterns of RON95 fuel prices in Malaysia using time series data.



Model Selection

To identify an appropriate ARIMA model for forecasting future RON95 fuel prices based on historical observations.



Performance

To examine the accuracy and effectiveness of the ARIMA forecasting model in predicting future RON95 fuel prices.

DATA DESCRIPTION

Source : Official Malaysia Open Data portal (data.gov.my)

Variables : Date, RON95, RON97, Diesel, Diesel East Malaysia

Variables of Interest : RON95, recorded weekly

Size : 464 weekly observations after cleaning

Technological Stack

Phyton (Jupyter Notebook) - Used for data exploration, data cleaning, ARIMA model and forecasting.

Power BI – Interactive Dashboard Development

METHODOLOGY

Preprocessing
Data Cleaning, filtering, date time indexing, and invalid entry removal.

Partitioning
80% Training set, 20% Testing test split.

Estimation
ACF/PACF examination for model identification

Analysis
Summary statistics calculated to provide an overview

Testing
Stationary testing via ADF & visual inspection

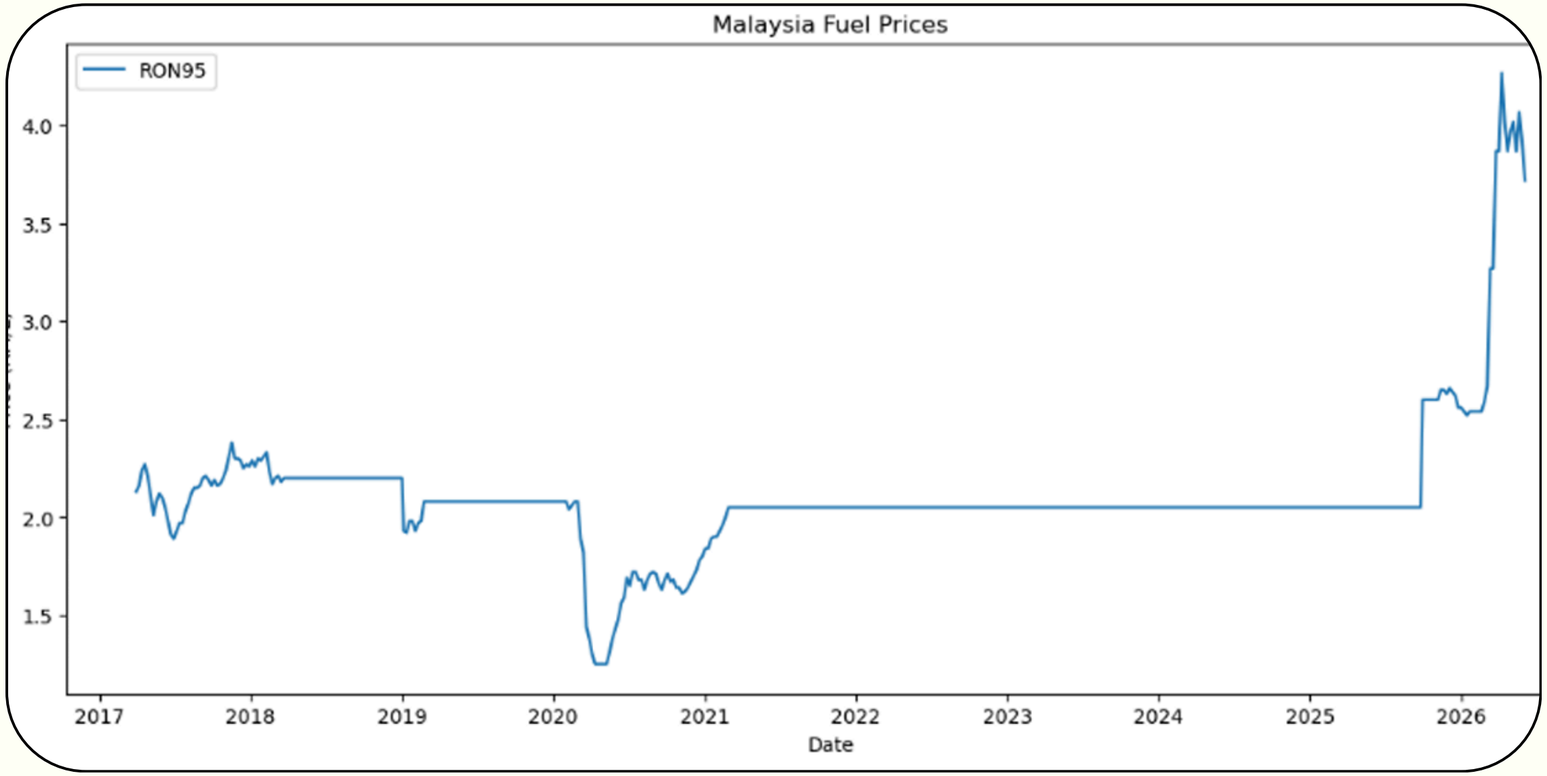
Evaluation
MAPE, RMSE & MAE accuracy measurement

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RESULTS

Results and discussion.

TIME SERIES PLOT AND DECOMPOSITION



The RON95 price series exhibits periods of fluctuation and stability.

- Prices fell sharply in 2020.
- Remained constant from 2021 to 2025.
- Increased significantly in 2026.

The time series does not appear to have a strong long-term upward or downward trend

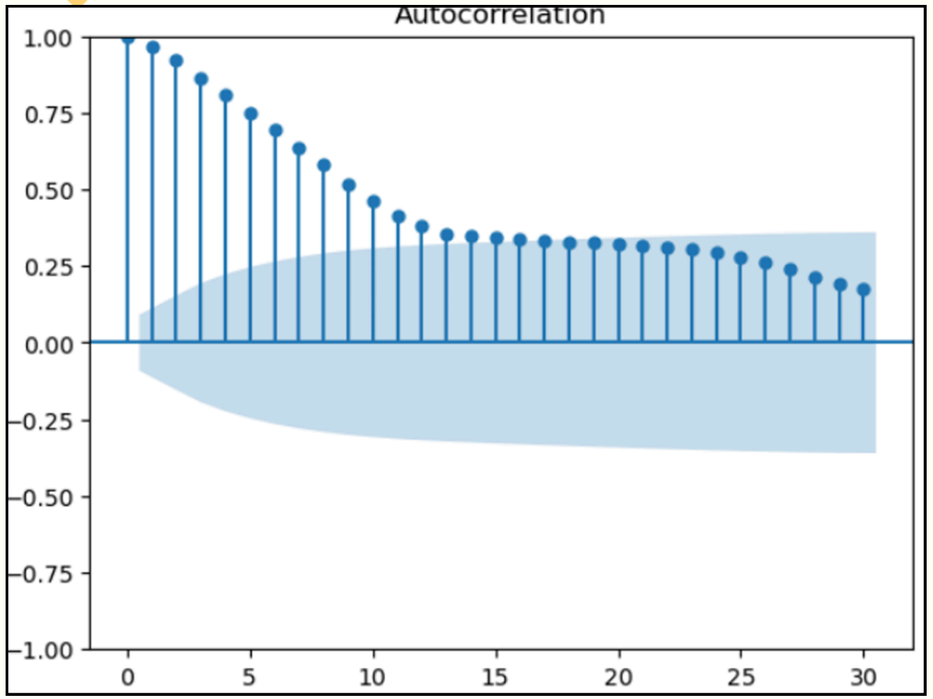
STATIONARITY TEST RESULTS

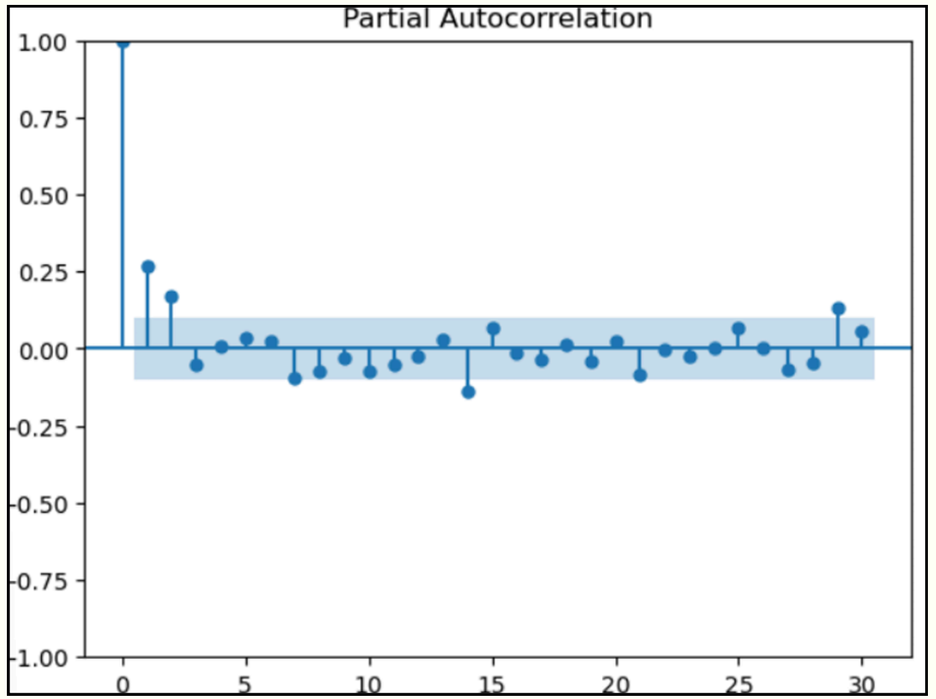
ADF Statistic: -3.154 p-value: 0.02278
As the p-value is below 0.05, the null hypothesis of a unit root is rejected, indicating that the original series is **stationary** and does **not require differencing.**

Therefore, d = 0

```
 RON95 Training
ADF Statistic: -3.154451264459925
p-value: 0.0227773014623027 7
Stationary
```

Augmented Dickey-Fuller (ADF)





Autocorrelation (ACF)

- The first two vertical bars (Lag 1 and Lag 2) cut through the blue confidence boundary line.
- Indicates that today's price is heavily influenced from the past 2 weeks.
- Hence, the second-order moving average component, MA(2) is appropriate.

Partial Autocorrelation (PACF)

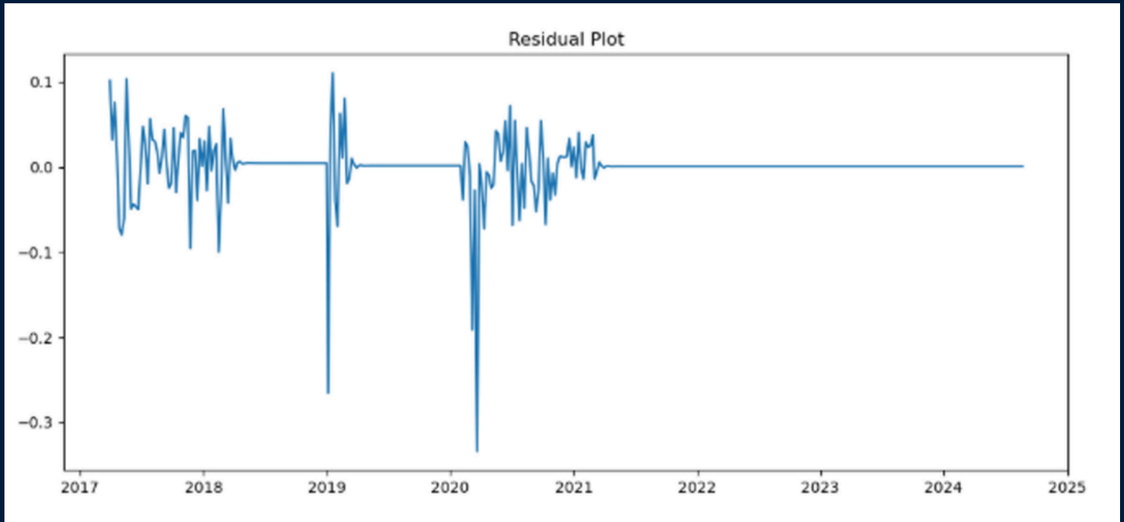
- The PACF plot displays a significant spike at lag 1.
- This indicates that the first lag has a strong direct effect on the current value of the series.
- Hence, the first-order autoregressive component, AR(1) is appropriate.

ARIMA MODEL SELECTION

ARIMA(1,0,2) was selected because it produced the lowest AIC value, indicating that it provides the best balance between model accuracy and efficiency.

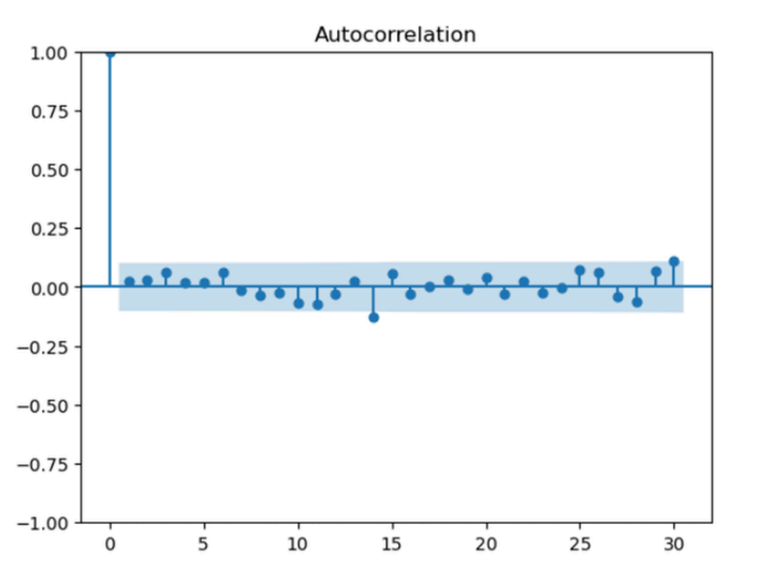


DIAGNOSTIC CHECKING



Diagnostic Plot Residual Over Time

- The residuals fluctuate randomly around zero with no clear pattern
- ARIMA model has successfully captured the underlying structure



Diagnostics Plots Acf of Residual

- Most residual autocorrelations fall within the confidence bounds
- Indicates the residuals resemble white noise and the model adequately explained the serial dependence

DIAGNOSTIC CHECKING

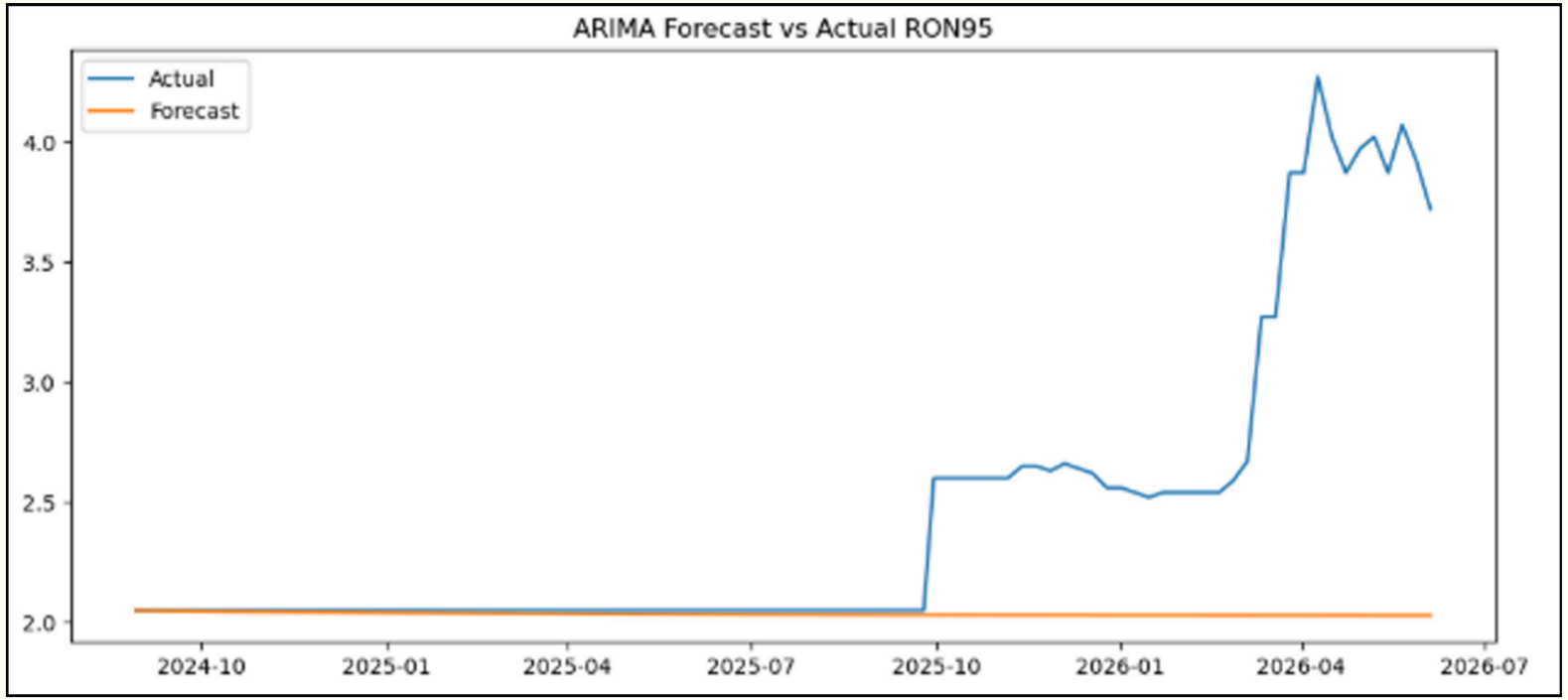
- P-value of 0.783055 (>0.05)
- H0 cannot be rejected, indicates residuals behave as white noise.
- ARIMA model adequately captures the underlying pattern of the fuel price series and is suitable for forecasting

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(371,)
     lb_stat  lb_pvalue
10  6.372522   0.783055
```

Figure 10 : Ljung-Box Test

Ljung-Box Test

FORECAST RESULTS AND ANALYSIS



12.28%
MAPE

0.40
MAE

0.74
RMSE

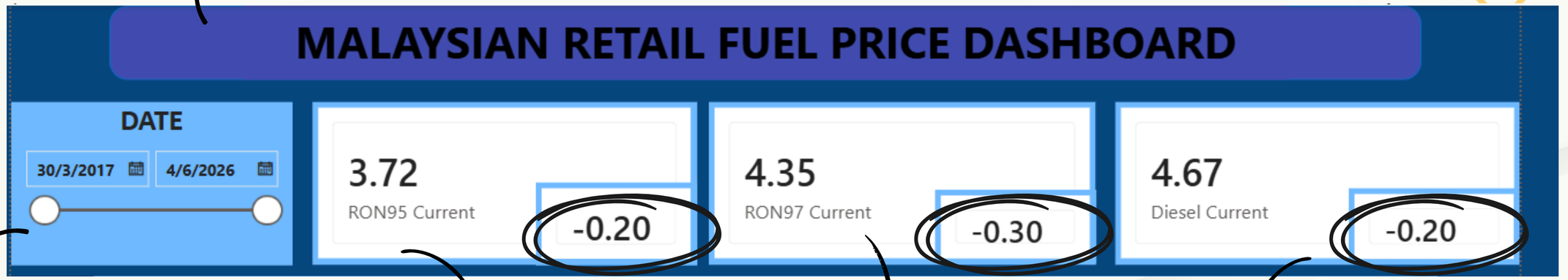
These results indicate that the model provides reasonably accurate forecasts for RON95 prices. MAPE value between 10% and 20% indicates good forecasting performance.

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DASHBOARD

Interactive Dashboard.

Title for the dashboard

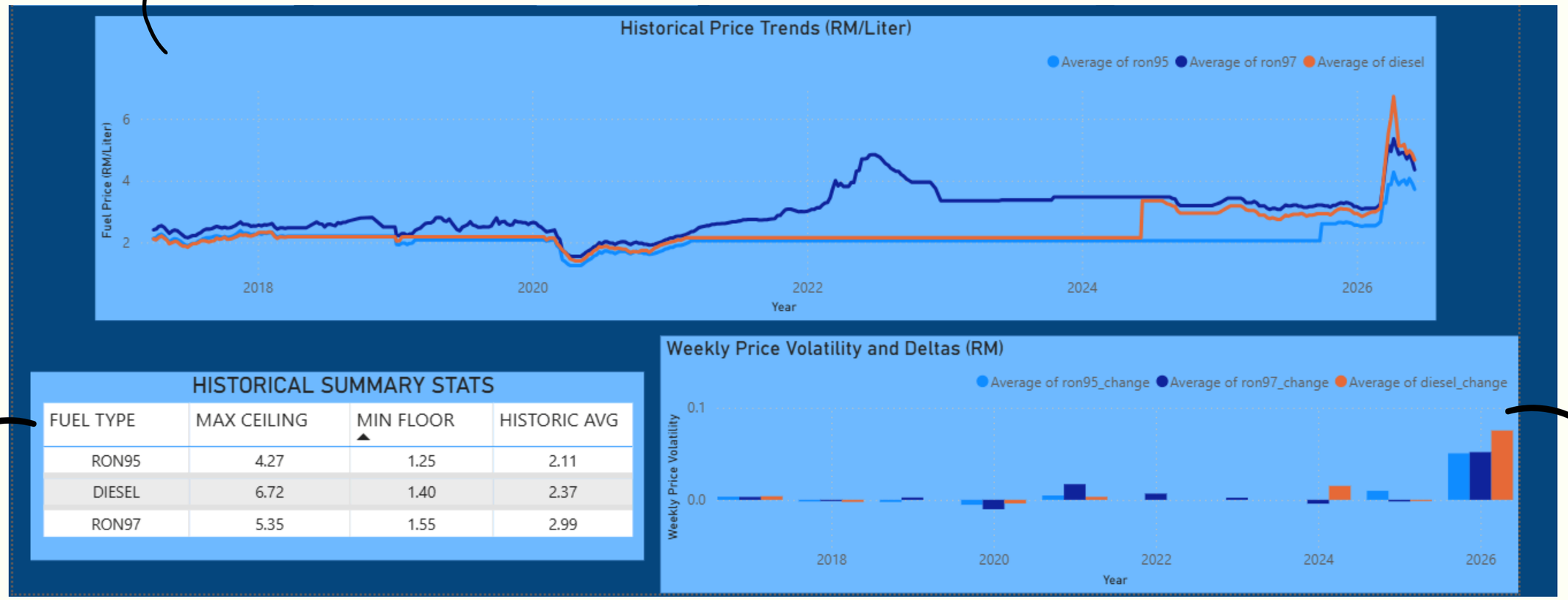


Date Filter - allows users to select specific period and analyze fuel price trends within the chosen timeframe

Current Fuel Price KPIs - display the latest recorded prices for RON95, RON97 and Diesel, as well as the difference from the previous reporting period.



Historical Price Trends (RM/Liter) - illustrates the movement of fuel prices over time, enabling users to identify long-term trends, price spikes and periods of stability.



Historical Summary Stats - provides descriptive statistics for each fuel type, including the maximum recorded price, minimum recorded price and historical average price.

Weekly Price Volatility and Deltas (RM) - shows weekly changes in fuel prices, helping users identify periods of high volatility and significant market adjustments.

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THANK YOU

Thank you for your attention.